

1. Define True/False. If false, give correct answer.10
- i. An interface in Java cannot inherit more than one interface.

ii. Abstract class can have overloaded constructor.

iii. Interfaces don't have instance variable.

iv. All methods in an *interface* must be static.

v. Super class variable can reference subclass object.

vi. Default members of a class can be accessed from subclass of different package.

vii. In java, a class can implement more than one interface.

viii. Whether you write your own constructor or not, JVM always provides a default constructor for your class.

ix. Interfaces can have abstract constructor.

x. abstract class can have only abstract methods.

2. What is the output of the following codes? Give explanation to each: (3+2) * 4

i)

<pre>public class Emp { static int bankVault; } public class TestDemo { public static void main(String args[]){ Emp emp1 = new Emp(); Emp emp2 = new Emp(); emp1.bankVault = 10; emp2.bankVault = 20; System.out.println(""+emp1.bankVault); System.out.println(""+emp2.bankVault); System.out.println(""+Emp.bankVault); } } public class Emp { static int bankVault; }</pre>	Ans:
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ii)

<pre>public class TestJava { void show(){ try{ String s = null; System.out.println("1"); } try{ System.out.println(s.length()); } catch (NullPointerException e){</pre>	a) 1 inner 2 outer b) 1 inner 2 c)
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<pre> System.out.println("inner"); } System.out.println("2"); } catch (NullPointerException e){ System.out.println("outer"); } } } public static void main(String args[]){ TestJava ob = new TestJava(); ob.show(); } }</pre>	1 inner outer d) Runtime Exception.
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iii)

<pre>class Animal{ void jump(){ System.out.println("Animal"); } } class Cat extends Animal{ void jump(int a){ System.out.println("Cat"); } } class Rabbit extends Animal{ void jump(){ System.out.println("Rabbit"); } } public class Circus { public static void main(String args[]){ Animal cat = new Cat(); Rabbit rabbit = new Rabbit(); cat.jump(); rabbit.jump(); } }</pre>	a) Cat Rabbit b) Animal Anima c) Animal Rabbit d) Compilation Error e) None of the above
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iv)

<pre>1. package A; 2. public class Foo { 3. int a=5; 4. protected int b = 6; 5. public int c=7; 6. } 7. package A.B; 8. import A.Foo; 9. public class Bar { 10. public static void main(String[] args){ 11. Foo f = new Foo(); 12. System.out.println(""+f.a); 13. System.out.println(""+f.b); 14. System.out.println(""+f.c); 15. } 16.}</pre>	a) 5 6 7 b) 5 6 7 c) Compilation fails with an error on line 11 d) Compilation fails with an error on line 12 e) Compilation fails with an error on line 13 f) Compilation fails with an error on line 14
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